ENVIRONMENTA

Community Medicine (PSM) | 3rd Year

Hi Sammy!
Environment = Health!
Swachh Bharat to
water quality — all here!

Environmental Health | PSM | 3rd Year | MBBS

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Water purification is THE most asked topic in environmental health!

SAFE WATER (WHO)

Free from pathogenic organisms, chemical & radioactive substances, pleasant to taste, usable for domestic purposes.

Purification of Water — At a Glance

Level	Methods	Key Points
Large Scale	Sedimentation	Plain — removes large particles; Coagulation — alum
	Filtration	Slow sand filter (best for rural) vs Rapid sand filter
	Disinfection	Chlorination — most widely used method globally
Household	Boiling	Most reliable method; kills all pathogens in 10-20 min
	Chlorination	Bleaching powder 2.5 g/L for well; sodium hypochlorite
	Filtration	Candle filter (Berkefeld), ceramic filters
Emergency	Chlorine tablets	Halazone tablets — 1 tab/1L water, wait 30 min

Bacteriological Standards of Water (WHO/IS:10500)

Parameter	Standard	Test Method
Coliform count	< 10/100 mL (treated)	MPN (Most Probable Number)
E. coli	0/100 mL (must be absent)	Membrane filter technique
Residual chlorine	0.5 mg/L (free chlorine)	OT test / Starch iodide test
pH	6.5 - 8.5	pH meter / indicator paper
Turbidity	< 1 NTU (ideal)	Nephelometer

Slow Sand Filter vs Rapid Sand Filter

Feature	Slow Sand Filter	Rapid Sand Filter
Rate of filtration	100-200 L/m2/hr	3000-6000 L/m2/hr
Biological layer	Vital Schmutzdecke present	Absent — purely mechanical
Bacteria removal	98-99% (best biological)	90-95% (needs disinfection)
Maintenance	Simple, low cost	Complex, backwashing needed
Best for	Rural India, small towns	Urban, large-scale plants

Horrock's apparatus and break-point chlorination are VIVA favourites!

Types of Chlorination

Type	Details
Free residual	Dose gives 0.5 mg/L free Cl after 1 hr — STANDARD method
Plain / Simple	0.5 mg/L added to water without demand testing
Double chlorination	Pre + post filtration (highly polluted water)
Super chlorination	5-10 mg/L — for epidemics, followed by dechlorination
Break-point	Chlorine added until free residual appears after demand met

★ **BREAK-POINT CHLORINATION (Exam Favourite!)**

1. Phase 1: Cl_2 destroys reducing substances (H_2S, Fe, Mn) — no residual
2. Phase 2: Cl_2 reacts with NH_3 — forms chloramines (combined residual)
3. Phase 3: Chloramines destroyed — dip = BREAK POINT — free Cl_2 appears!

Horrock's Apparatus (for Well Chlorination)

Purpose:	To determine the dose of bleaching powder needed for a well
Principle:	Add increasing doses of Cl to 6 cups (200 mL each) of well water
Indicator:	Starch-iodide paper turns BLUE = adequate residual chlorine
Contact time:	1 hour after adding bleaching powder
Pass:	Cup showing blue colour after 1 hr = correct dose for that well

Tests for Chlorine in Water

Test	Detects	Colour Change
OT (Ortho-tolidine)	Free + combined Cl	Yellow = adequate (>0.2 mg/L)
Starch-iodide	Free chlorine	Blue colour = chlorine present
DPD test	Free Cl (most accurate)	Pink colour = free Cl present

★ **Memory: OT = Old Test, DPD = Developed Precise Detection**

OT is being replaced by DPD in modern labs. But OT is still asked in exams!

SPM, SO2, CO effects
and sewage treatment
are exam regulars!

Major Air Pollutants + Health Effects

Pollutant	Source	Health Effect	Standard (NAAQS)
SPM/PM10	Vehicles, industry, dust	Resp disease, cancer	100 ug/m3 (24 hr)
PM2.5	Combustion, diesel	Deep lung, CVS risk	60 ug/m3 (24 hr)
SO2	Fossil fuels, industry	Bronchitis, acid rain	80 ug/m3 (24 hr)
CO	Vehicles, incomplete burn	COHb — headache, death	2 mg/m3 (8 hr)
NO2	Vehicles, power plants	Resp irritation	80 ug/m3 (24 hr)
O3 (ozone)	Photochemical smog	Lung damage, wheeze	100 ug/m3 (8 hr)
Lead	Leaded petrol, paint	Neuro, renal toxicity	0.5 ug/m3 (annual)

★ INDOOR AIR POLLUTION (Indian Context)

Biomass fuel (wood, dung) used by 70%+ rural India = CO, PM, formaldehyde

Ujjwala Yojana (2016) — free LPG to BPL families — reduces indoor pollution

Sewage Treatment — Steps

Stage	Process	What Happens
Preliminary	Screening, grit removal	Removes large solids, sand, gravel
Primary	Sedimentation tank	Settles suspended solids (50-70% removed)
Secondary	Biological treatment	Activated sludge / Trickling filter — BOD removal
Tertiary	Advanced treatment	Removes N, P, pathogens — for reuse
Disinfection	Chlorination / UV	Kills remaining pathogens before discharge

★ KEY TERMS:

BOD (Biochemical Oxygen Demand) = O2 needed to decompose organic matter — lower = cleaner

COD (Chemical Oxygen Demand) = total O2 for all oxidisable matter — always > BOD

Swachh Bharat and Jal Jeevan Mission are HOT topics for PSM exams!

Standards of Housing (Park's)

Parameter	Standard / Recommendation
Floor space/person	50-100 sq ft per person (minimum)
Cubic space/person	300-500 cu ft (ventilation requirement)
Lighting	1/10th of floor area should be window area
Ventilation	Cross-ventilation; 2-3 air changes/hour (natural)
Overcrowding	>2 persons per room (Census definition)
Water supply	135 L/person/day (urban) — IS standard
Waste disposal	Sanitary latrine — water-seal type recommended

Swachh Bharat Mission (SBM)

Launched:	2nd October 2014 (Gandhi Jayanti) by PM Modi
Goal:	Open Defecation Free (ODF) India by 2nd Oct 2019
SBM-Gramin:	Rural — toilet construction, behaviour change, SLWM
SBM-Urban:	Urban — toilets + solid/liquid waste management
SBM 2.0:	Launched 2020 — ODF Plus = ODF sustainability + waste mgmt

Jal Jeevan Mission (JJM)

Launched:	2019 — Ministry of Jal Shakti
Goal:	Functional Household Tap Connection (FHTC) to every rural household
Standard:	55 litres per capita per day (lpcd) potable water
Target:	Har Ghar Jal by 2024 — piped water for all rural homes

Solid Waste Management Methods

Method	Principle	Suitability
Sanitary landfill	Burial + compaction + cover	Most common urban method in India
Composting	Biological decomposition	Organic waste — manure production
Incineration	High-temp burning	Biomedical waste, hazardous waste
Vermiculture	Earthworm-based composting	Small-scale, organic waste
Recycling	Reuse of materials	Paper, plastic, glass, metal

Environmental health = easy marks if you know the tables! Go Sammy!

Most Repeated LAQs (10–12 marks)

- 1. Describe purification of water on a large scale with diagram
- 2. Write about chlorination of water — types, break-point chlorination
- 3. Horrock's apparatus — principle, procedure, and interpretation
- 4. Describe sewage treatment steps. What is activated sludge process?
- 5. Write about Swachh Bharat Mission — objectives, components, achievements

Most Repeated SAQs (5–6 marks)

- 1. Slow sand filter vs rapid sand filter (table)
- 2. Break-point chlorination with graph
- 3. Horrock's apparatus — procedure
- 4. Standards of drinking water (bacteriological)
- 5. Jal Jeevan Mission — objectives and features
- 6. BOD and COD — define and significance

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Most reliable method of water purification	Boiling
Most widely used disinfectant for water	Chlorine
Best biological filter	Slow sand filter (Schmutzdecke)
Chlorine demand test for wells	Horrock's apparatus
Residual chlorine for safe water	0.5 mg/L free chlorine
OT test colour for adequate Cl	Yellow
Break-point = what appears?	Free residual chlorine
BOD of clean water	< 2 mg/L
Most common air pollutant indicator	SPM / PM10
Indoor air pollution fuel — India	Biomass (wood, dung, crop residue)
Ujjwala Yojana provides	Free LPG to BPL families
JJM target supply	55 lpcd piped water to all rural homes

♥ **Clean water = healthy nation, Sammy! You'll help make it happen!**
This topic is your scoring machine in PSM. Own it!